

AI BASED NEURO- KINETIC AERIAL INTERCEPTION

Wildlife Emergency Response Ops

AI-powered aerial intelligence for real-time threat assessment and safer human-wildlife emergency response.

THE TEAM BEHIND AIRAVATA

Four minds. One system. A safer way to respond.

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WHEN HUMAN–WILDLIFE CONFLICT BECOMES AN **EMERGENCY**

The danger is not only the animal. It's the time between detection and safe intervention.

01 EXPANDING RISK ZONES

Settlements increasingly intersect wildlife buffer zones, increasing the likelihood of dangerous encounters.

02 SLOW & DANGEROUS RESPONSE

Conventional patrols and static watchtowers can delay intervention while exposing responders to the danger zone.

03 UNCERTAIN TARGET ASSESSMENT

Visual estimation of animal size can lead to inaccurate intervention decisions, including ineffective or excessive dosing.

04 HUMAN EXPOSURE

Manual intervention under stress introduces response lag, precision challenges and additional risk to field teams.



THE RESPONSE WINDOW IS SHORT.
THE RISK IS NOT.

WHAT IF THE FIRST RESPONDER WASN'T A PERSON?

Airavata is an AI-assisted aerial emergency-response platform designed to sense, assess and support intervention before humans enter the danger zone.



01 SENSE

Detect the environment, target and movement.

02 UNDERSTAND

Build a real-time picture of the situation.

03 DECIDE

Assess risk and determine the appropriate response.

04 RESPOND

Support controlled intervention with humans in the loop.

The drone doesn't replace the responder. It changes where the responder begins.

HUMAN OVERSIGHT
● ACTIVE

AUTONOMOUS
PERCEPTION + NAVIGATION

CRITICAL ACTION
HUMAN AUTHORIZED

FROM DETECTION TO RESPONSE

One incident. Multiple intelligence layers. One controlled response.



PERCEPTION / ASSESSMENT | HUMAN AUTHORITY | CONTROLLED RESPONSE

AIRAVATA DOES NOT AUTOMATE THE HUMAN.
IT AUTOMATES THE INTELLIGENCE AROUND THE HUMAN.

THE MATHEMATICS BEHIND AIRAVATA

From 3D perception to a continuously updated target model.

1. Focusing & Gimbal Aiming at the Tiger's Thigh

To steer the optical/thermal gimbal to lock onto the rear thigh anatomical keypoint $\mathbf{p}_{\text{thigh}} = [u_{\text{thigh}}, v_{\text{thigh}}]^T$ identified by the pose model relative to camera principal point (c_x, c_y) and focal length (f_x, f_y) :

$$\begin{bmatrix} \Delta\theta_{\text{yaw}} \\ \Delta\theta_{\text{pitch}} \end{bmatrix} = \begin{bmatrix} \arctan\left(\frac{u_{\text{thigh}} - c_x}{f_x}\right) \\ \arctan\left(\frac{v_{\text{thigh}} - c_y}{f_y}\right) \end{bmatrix}$$

- $(u_{\text{thigh}}, v_{\text{thigh}})$: Pixel coordinates of the detected rear thigh keypoint.
- (c_x, c_y) : Image sensor optical center coordinates.
- (f_x, f_y) : Camera focal length components in pixels.

2. Neural Flight Steering (Sensorimotor Asymmetry)

To translate real-time motor imagery into continuous lateral velocity and yaw rate:

$$\mathbf{v}_{\text{cmd}} = K_v \tanh\left(\log_{10}\left(\frac{\text{Var}(\mathbf{w}_{C4}^T \mathbf{X})}{\text{Var}(\mathbf{w}_{C3}^T \mathbf{X})}\right)\right)$$

- $\mathbf{v}_{\text{cmd}}$: Commanded flight velocity vector $(v_y, \dot{\psi})$ for lateral drift and yaw rate.
- K_v : Flight velocity sensitivity gain matrix.
- $\mathbf{X}$: Multichannel EEG recording epoch.
- $\mathbf{w}_{C3}, \mathbf{w}_{C4}$: Common Spatial Pattern (CSP) spatial filters tuned to the contralateral hand areas (C_3).

3. The Mathematics of the Trajectory

The kinematic and aerodynamic equation governing the trajectory of a launched projectile (such as a dart or marker) under the effects of gravity, atmospheric drag, and crosswind is:

$$\mathbf{e}_{\text{proj}} = \mathbf{g} - \frac{\rho_{\text{air}} C_d A}{2m_{\text{proj}}} |\mathbf{v}_{\text{rel}}| |\mathbf{v}_{\text{rel}}|$$

- $\mathbf{a}_{\text{proj}}$: Instantaneous net acceleration vector of the projectile (m/s^2).
- $\mathbf{g}$: Acceleration vector due to gravity $(0, 0, -9.81)^T \text{ m/s}^2$.
- ρ_{air} : Local ambient air density (kg/m^3).
- C_d : Aerodynamic drag coefficient of the projectile.
- A : Frontal cross-sectional reference area of the projectile (m^2).
- m_{proj} : Total mass of the projectile including its payload (kg).
- $\mathbf{v}_{\text{rel}}$: Relative velocity vector between the projectile and ambient wind:

$$\mathbf{v}_{\text{rel}} = \mathbf{v}_{\text{proj}} - \mathbf{v}_{\text{wind}}$$

4. Projectile Trajectory & Ballistics

Each fired shell follows a ballistic trajectory, modeled using classical projectile motion:

$$V = \frac{\tilde{M} \cdot \left(2 \frac{\text{mg}}{\text{kg}} + 2 \frac{\text{mg}}{\text{kg}} \cdot S\right)}{C}$$

Where:

- V : Dispensed liquid drug volume (mL).
- $\hat{M}$: Estimated tiger body mass (kg)
- S : Sex indicator variable ($S = 1$ for male $\rightarrow 4 \text{ mg/kg}$, $S = 0$ for female $\rightarrow 2 \text{ mg/kg}$)
- C : Stock drug concentration (mg/mL, standard Zoletil 100 reconstituted at 100 mg/mL)

5. Projectile Dynamics & Control

The kinematic model for the projectile in flight (disregarding minor perturbations):

$$\mathbf{x}(t) = \mathbf{x}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2$$

Where:

- $\mathbf{x}(t)$: Position vector at time t .
- $\mathbf{x}_0$: Initial position vector.
- $\mathbf{v}_0$: Initial velocity vector.
- $\mathbf{a}$: Acceleration vector (includes gravity and drag components).

PERCEIVE → ESTIMATE → INTERPRET → PREDICT → COMMUNICATE → ACT
Every major subsystem is represented by a measurable mathematical model.

INTELLIGENCE ONBOARD. HUMAN CONTROL AT THE CRITICAL POINT.

Airavata uses shared autonomy to handle flight complexity while keeping critical decisions under human oversight.



NEURAL CONTROL STATION

- EEG / BCI
- NEURAL INPUT

AIRAVATA UAV

- PERCEPTION
- NAVIGATION
- SHARED AUTONOMY

TARGET TRACKING

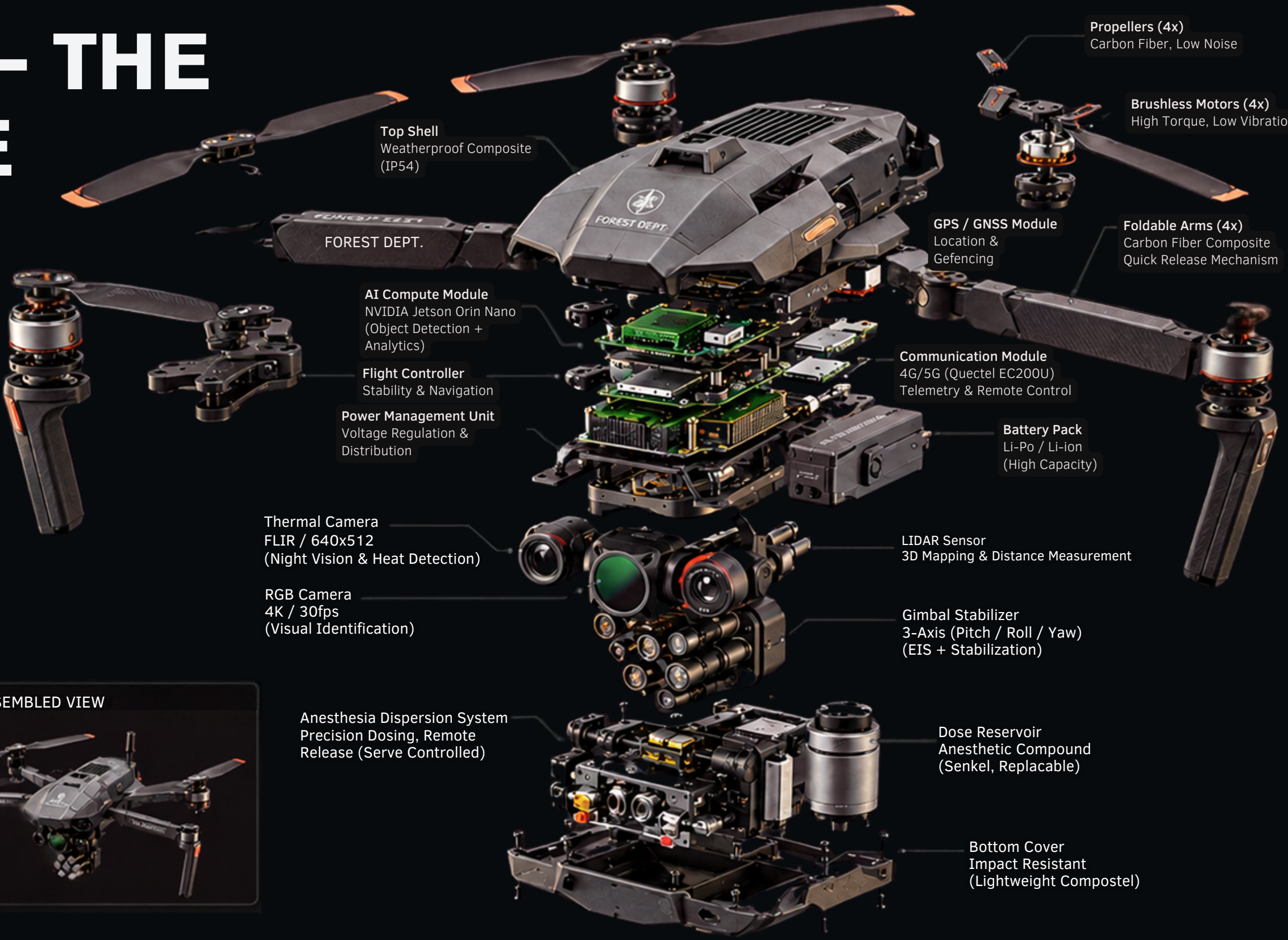
- VISION
- LiDAR
- AI ASSESSMENT

REMOTE OBSERVATION

- ENVIRONMENTAL SENSING

AUTONOMOUS WHERE SPEED MATTERS.
HUMAN-CONTROLLED WHERE CONSEQUENCE MATTERS.

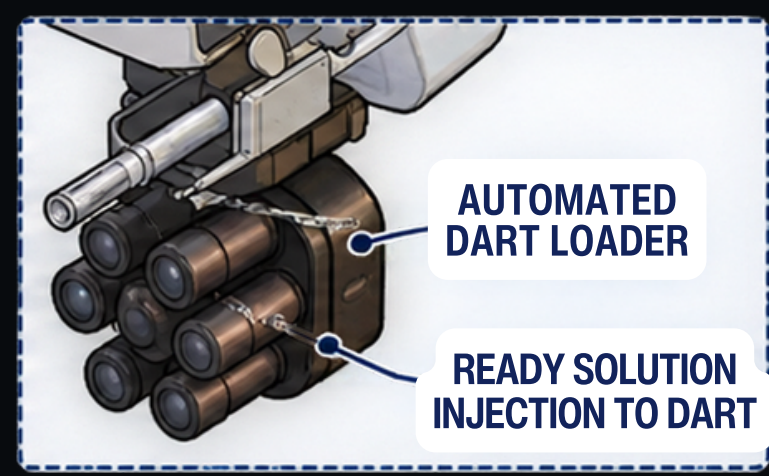
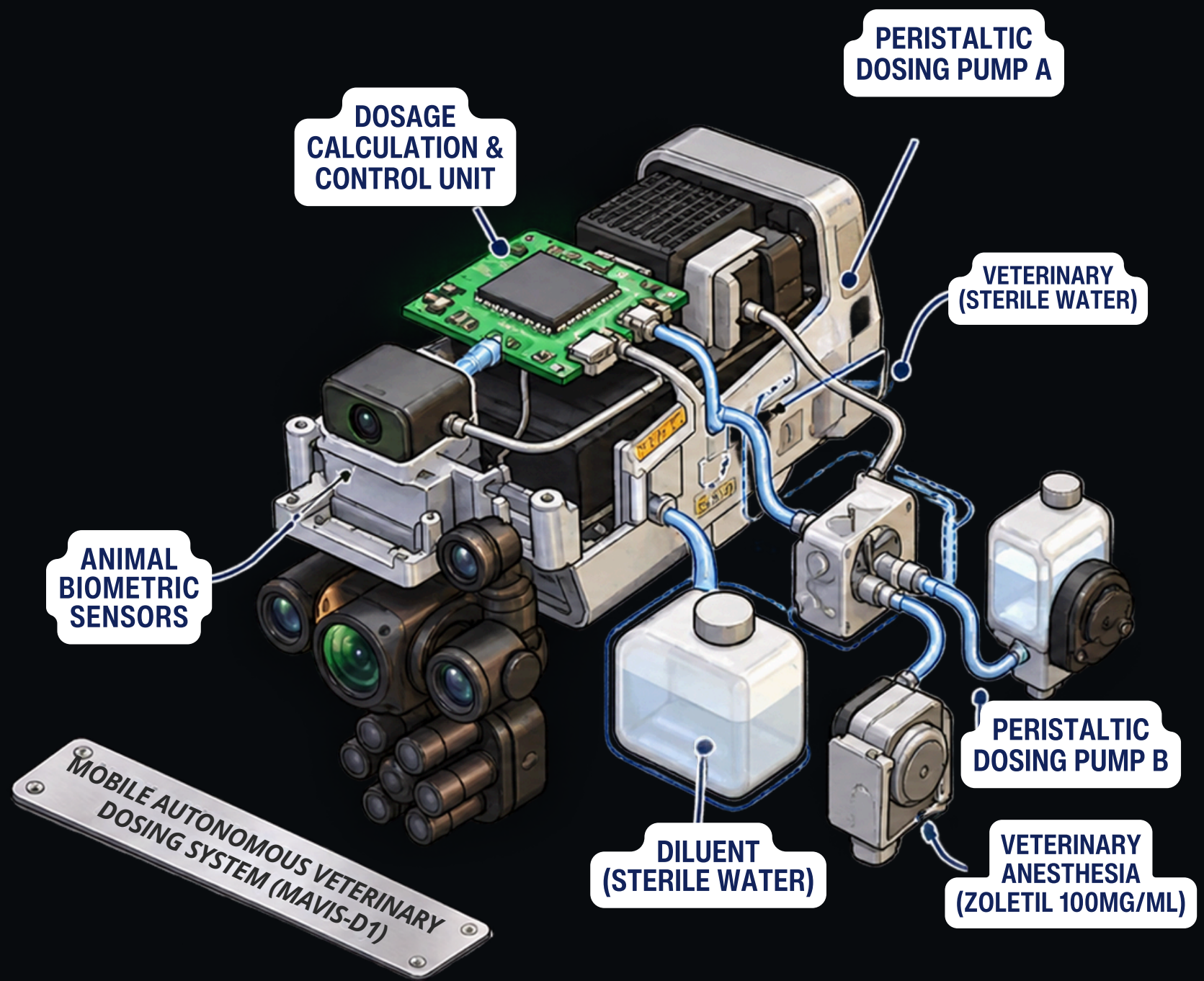
AIRAVATA — THE RESPONSE PLATFORM



- ### KEY FEATURES
- AI-Powered Animal Identification**
Spices, sex, height, weight estimation
 - Smart Anesthesia Dosing**
Based on AI + ML analysis
 - LIDAR Mapping**
Precise distance & terrain mapping
 - Thermal + RGB Vision**
Day / Night operations
 - Autonomous Navigation**
GPS + AI path planning
 - Real-time Telemetry**
4G/5G / Satellite (optional)
 - Tactical Deployment**
Rugged, field-ready design

- ### SYSTEMS BREAKDOWN
- Flight & Navigation**
Motors, ESC, FC, GPS, IMU
 - Sensing & Perception**
RGB, Thermal, IRM8, Sensors
 - AI & Compute**
Jetson Orin Nano, ML Models
 - Communication**
4G/5G, Remote Control
 - Power**
Battery, PDU, Regulators
 - Payload**
Anesthesia System, Reservoir
 - Structure**
Frame, Arms, Shell, Gimbal

THE INTERVENTION MECHANISM



AUTOMATIC DOSAGE FORMULATION

AI-Calculated Body Weight (kg) (from sensors)

4mg / kg (Pre-set Zoletil dose rate)

Total Zoletil (mg)

- Pumps command >> Calculated Vol. Zoletil + Vol. Diluent
- Mixing >> Ready Solution
- Output >> Calculated Volume for Dart

NEURAL CONTROL, TRANSLATED INTO FLIGHT

Human intent is captured as a neural signal, interpreted by the control system, and translated into high-level commands for Airavata.



WIRELESS TRANSMISSION



NEURO-BCI SIGNAL MAPPING

- FZ, F3, F4 – PREFRONTAL: TBR & ALPHA ASYMMETRY → FOCUS/STRESS → SAFETY ABORT
- C3, C4 – MOTOR CORTEX: M-ERD (8–12 HZ) → LEFT/RIGHT STEERING & ARMING
- PZ – PARIETAL: P300 → TARGET-ATTENTION CONFIRMATION
- O1, O2 – OCCIPITAL: ALPHA SUPPRESSION → VISUAL ENGAGEMENT

(10) Publication No: **IN202621056313 A1**

(22) Date of Filing: 03-05-2026



(19) Intellectual Property Office, India

(43) Publication Date: 03-07-2026

Journal No: 27/2026

(12) INDIAN PATENT APPLICATION

(54) Title: **Ai automated animal detection and neutralizing system**

(51) International Classification:

**A61D 7/04, A61D 7/00, A61B 5/00, A61D 17/00
A61M 16/01**

(21) Application No: **202621056313**

(31) Priority Document No: --

(32) Priority Date: --

(86) International Application No: --
Filing Date: --

(87) International Publication No: --

(61) Patent of Addition to Application No: --
Filing Date: --

(62) Divisional to Application No: --
Filing Date: --

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SECURED THE FUNDING OF 1.75 LAKHS FROM CSE DEPARTMENT ,YCCE

FROM PROTOTYPE TO DEPLOYMENT

A deployable response system for high-risk human–wildlife conflict zones.

BUSINESS MODEL

- **B2G + B2B Customers -**

1. Forest Departments
2. Animal Reserves & Wildlife Authorities
3. Conservation Organizations
4. CSR / Environmental Programs

Revenue -

Hardware Deployment + Installation +
Annual Software Subscription

VALIDATION

- **Prototype** : Ready
- **Pilot users**: AI Detection + Neural Steering + RE-ID + Buffer Monitoring + Admin Dashboard
- **Pilot Target** : 1 animal reserve
- **Next milestone**: Controlled field pilot

TRACTION

- Revenue: Pre-revenue
- Stage: Prototype → Field Pilot
- Current progress:
- Working AI detection concept
- Connected hardware architecture
- Web-based admin control system
- Human-in-the-loop response workflow
- Key KPI:
- Successful conflict-prevention / deterrence rate

FROM CONTROLLED PROTOTYPE TO FIELD DEPLOYMENT

A staged path from system integration to validated operations.

OUR ASK

WHAT WE ARE SEEKING

PILOT ACCESS

Selected animal reserves & buffer zones

MENTORSHIP

Wildlife + AI + Drone Technology + field deployment experts

PARTNERSHIPS

Forest departments & conservation organizations

FUNDING

Hardware development, deployment & field validation

MARKET

- TAM: ₹14,500 – ₹18,000 Crore (\$1.75B – \$2.15B USD)
- SAM: ₹1,800 – ₹2,400 Crore (\$215M – \$290M USD)
- SOM: ₹300 – ₹450 Crore (\$36M – \$54M USD)

IMPACT

- **COMMUNITIES**
Earlier detection → More response time → Reduced human risk
- **WILDLIFE**
Early intervention → Fewer dangerous encounters → Safer coexistence
- **FOREST AUTHORITIES**
Real-time intelligence → Faster decisions → Better field response
- **CONSERVATION**
Individual animal data → Movement patterns → Better wildlife management
- **SDGs -**
1. **SDG 11** — Sustainable Communities
2. **SDG 15** — Life on Land

DON'T SEND PEOPLE INTO THE UNKNOWN.

SEND INTELLIGENCE FIRST.

AI-powered aerial intelligence for safer human–wildlife emergency response.

LAT 21.1458° N
LON 79.0882° E
ALT 487 m
SPD 12.4 m/s
BAT 78%

THANK YOU

TEAM AIRAVATA
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AI PERCEPTION / HUMAN AUTHORIZATION / SHARED AUTONOMY